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Assigment: Advanced DB TAsk 1

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1. Database Creation And Inserting Data

Sunday 19:15

File Edit Selection smart_home

create-db-template.sql

home > martell0x1 > .dbclient > storage > 1770129510602@@127.0.0.1@1433 > create-db-template.sql > ...

```sql
CREATE DATABASE CollegeRegistrationDB;  1s
```

Result

Search Results Cost: 1s

Execution completed in 1s

Sunday 19:17

File Edit Selection smart_home

create-db-template.sql

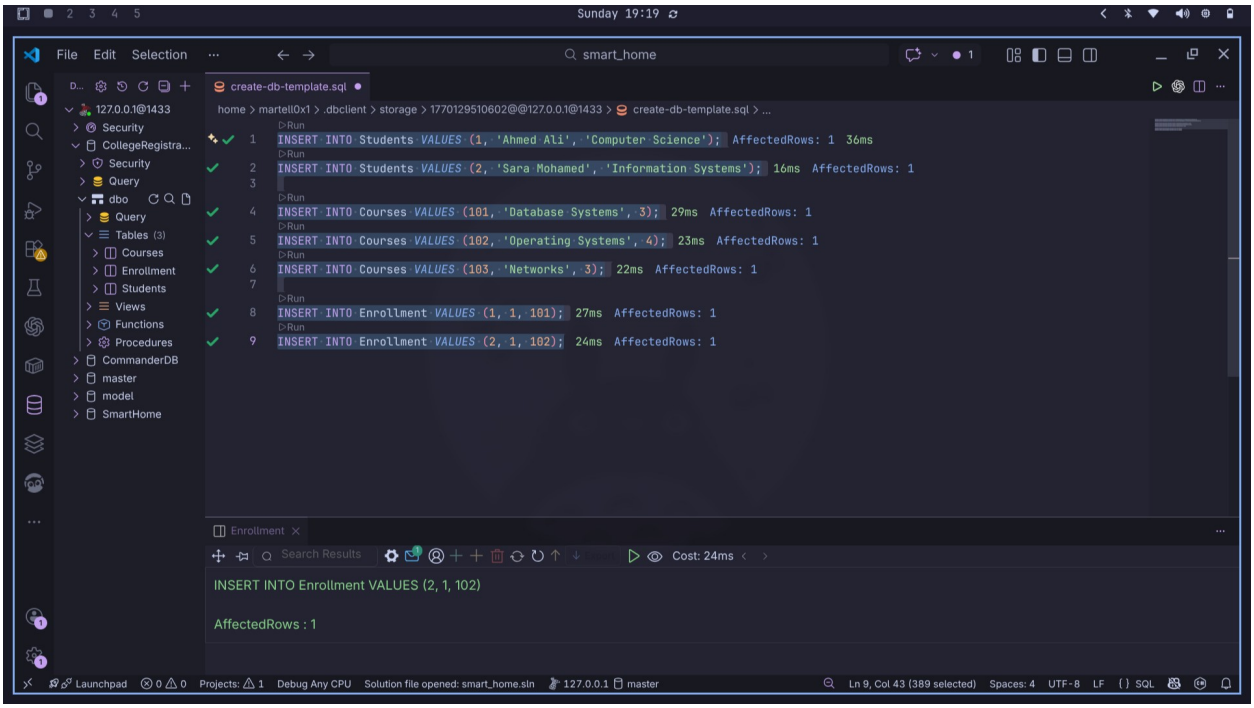
home > martell0x1 > .dbclient > storage > 1770129510602@@127.0.0.1@1433 > create-db-template.sql > ...

```sql
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    student_name VARCHAR(100) NOT NULL,
    major VARCHAR(100) NOT NULL
);  40ms

CREATE TABLE Courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(100) NOT NULL,
    credit_hours INT NOT NULL
);  29ms

CREATE TABLE Enrollment (
    enrollment_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    CONSTRAINT FK_Student FOREIGN KEY (student_id)
        REFERENCES Students(student_id),
    CONSTRAINT FK_Course FOREIGN KEY (course_id)
        REFERENCES Courses(course_id)
);  33ms
```

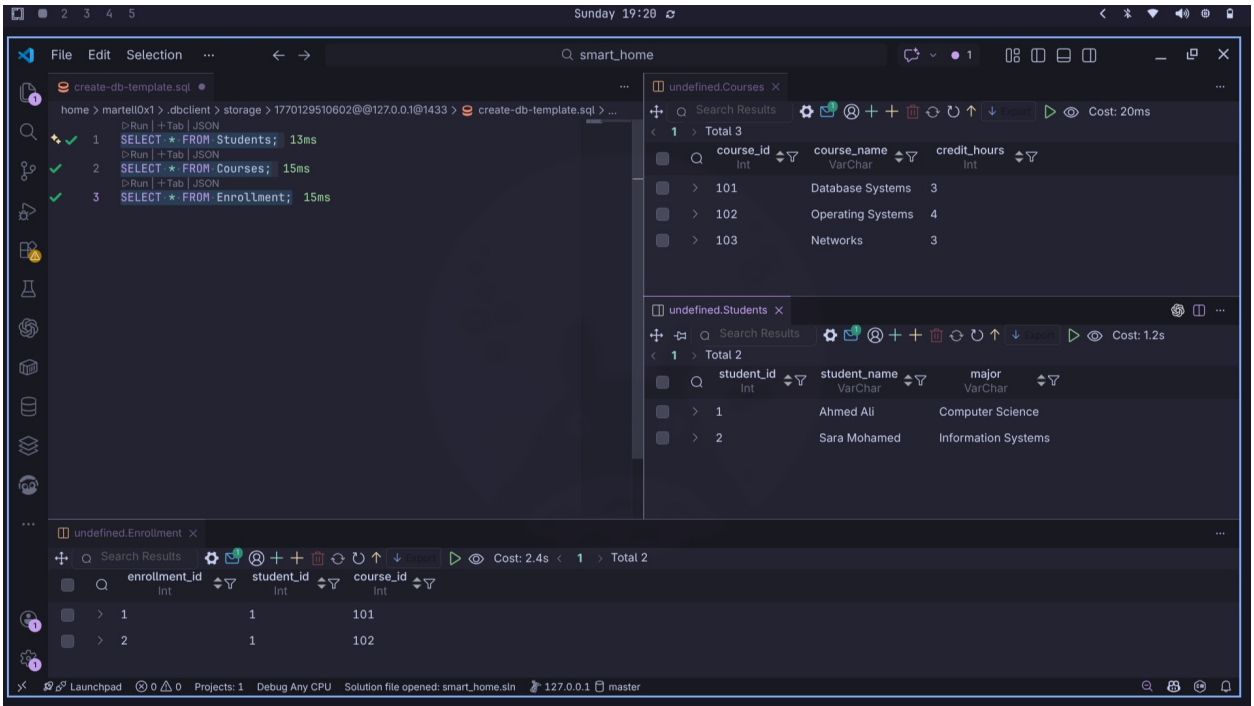
Result

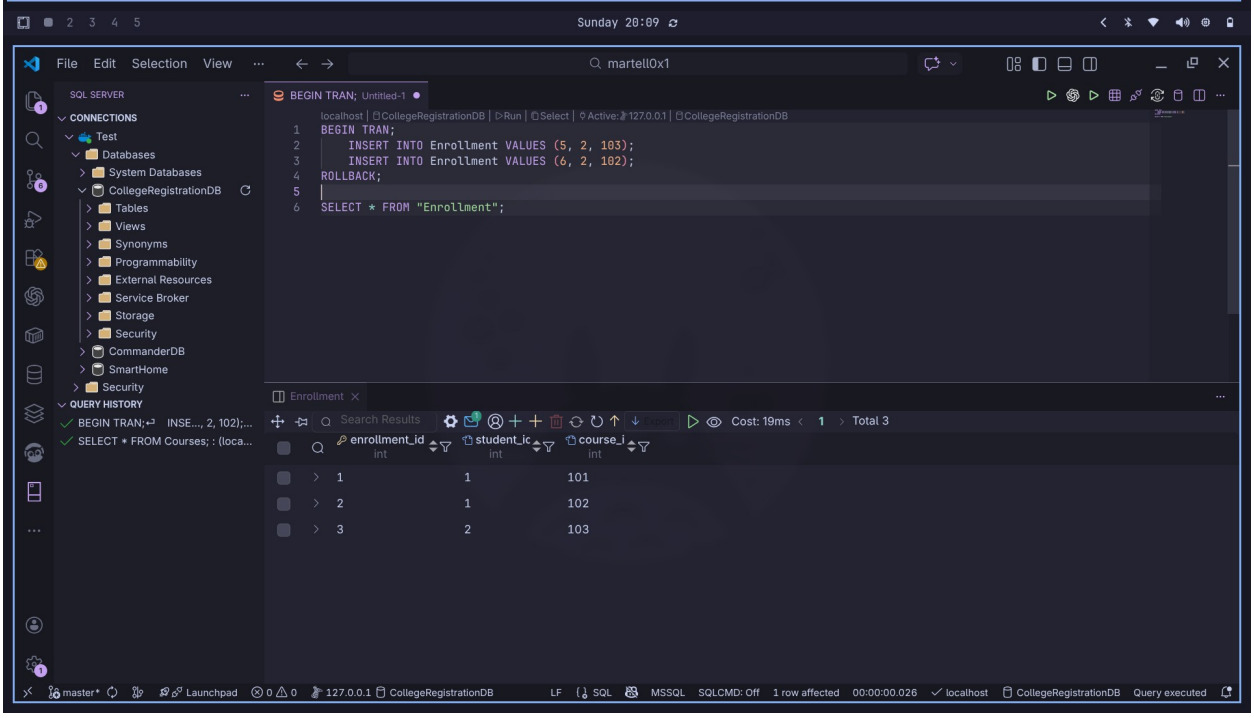
Search Results Cost: 33ms

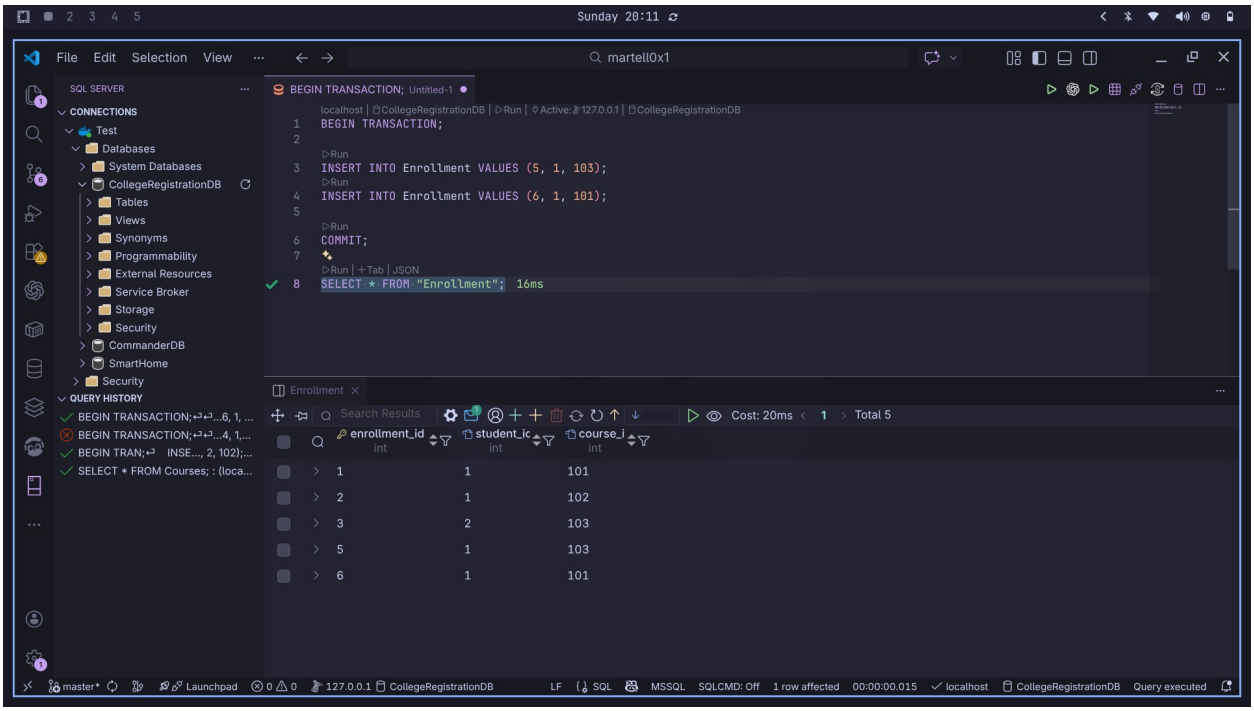
Execution completed in 33ms



2. Viewing Tables after and before Transaction


Sunday 19:20
File Edit Selection
smart_home
create-db-template.sql
home > martell0x1 > .dbclient > storage > 1770129510602@@127.0.0.1@1433 > create-db-template.sql > ...
Run | +Tab | JSON
1 SELECT * FROM Students; 13ms
2 SELECT * FROM Courses; 15ms
3 SELECT * FROM Enrollment; 15ms
undefined.Courses
Search Results
Cost: 20ms
Total 3
course_id Int
course_name VarChar
credit_hours Int
101 Database Systems 3
102 Operating Systems 4
103 Networks 3
undefined.Students
Cost: 1.2s
Total 2
student_id Int
student_name VarChar
major VarChar
1 Ahmed Ali Computer Science
2 Sara Mohamed Information Systems
undefined.Enrollment
Cost: 2.4s
Total 2
enrollment_id Int
student_id Int
course_id Int
1 1 101
2 1 102
Launchpad
Projects: 1
Debug Any CPU
Solution file opened: smart_home.sln
127.0.0.1
master


Sunday 20:09
File Edit Selection View
martell0x1
SQL SERVER
CONNECTIONS
Test
Databases
System Databases
CollegeRegistrationDB
Tables
Views
Synonyms
Programmability
External Resources
Service Broker
Storage
Security
CommanderDB
SmartHome
Security
QUERY HISTORY
BEGIN TRAN; INSE..., 2, 102);...
SELECT * FROM Courses; : (loca...
BEGIN TRAN; Untitled-1
localhost | CollegeRegistrationDB | Run | Select | Active: 127.0.0.1 | CollegeRegistrationDB
BEGIN TRAN;
INSERT INTO Enrollment VALUES (5, 2, 103);
INSERT INTO Enrollment VALUES (6, 2, 102);
ROLLBACK;
SELECT * FROM "Enrollment";
Enrollment
Search Results
Cost: 19ms
Total 3
enrollment_id int
student_ic int
course_i int
1 1 101
2 1 102
3 2 103
master*
Launchpad
127.0.0.1
CollegeRegistrationDB
LF
SQL
MSSQL
SQLCMD: Off
1 row affected
00:00:00.026
localhost
CollegeRegistrationDB
Query executed



Questions:

1. What is the difference between COMMIT and ROLLBACK?

-> COMMIT saves changes after transaction occures in the DBMS

-> ROLLBACK undo changes and returns the data state into the previous state before transaction

2. What problems could occur if transactions were not used?

-> If transactions aren't used, data can become **inconsistent** when only part of an operation succeeds.
Partial updates may cause **data loss** if a failure occurs mid-operation.
Concurrent operations can lead to **race conditions**, where simultaneous changes corrupt the data.

3. Which ACID property is demonstrated when using ROLLBACK?

-> Atomicity